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Research Paper

Integrated chromosome-level assembly and resequencing uncover the genetic basis of fruit size differentiation in Russian olive

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ABSTRACT

Russian olive, which is widely distributed in arid and semiarid regions of Central Asia, is an economically and ecologically significant tree species. Over long-term evolution, Russian olive has differentiated into two types: the large-fruited form (*Elaeagnus moorcroftii*) and the small-fruited form (*E. oxycarpa or E. angustifolia*). However, our understanding of the molecular regulation of fruit size in Russian olive has been limited. In this study, by employing a comprehensive approach using Oxford Nanopore Technologies, Illumina sequencing, and high-throughput chromosome conformation capture (Hi-C), we report a chromosome-level genome assembly of *E. moorcroftii*, spanning 505 Mb (contig N50 = 27.42 Mb), and containing 33,650 protein-coding genes. We subsequently resequenced the genomes of 70 accessions from Xinjiang. Selective sweep analysis revealed 1444 genes associated with genetic differences between large- and small-fruited Russian olives. A total of 19 candidate genes associated with the regulation of fruit size were identified by combining GO enrichment analysis, including GATA transcription factor 1 (*GATA1*) and auxin-responsive protein IAA4-like (*IAA4*), among others. Four of these genes were further functionally validated via heterologous expression in transgenic *Arabidopsis thaliana*. These genes are speculated to play key roles in the regulation of fruit size. Collectively, our findings reveal the key regulatory genes underlying the genetic mechanism of fruit size in Russian olive and provide a theoretical basis for optimizing molecular breeding strategies.

1. Introduction

Russian olives, which belong to the Elaeagnaceae family, are deciduous trees that hold significant economic and ecological importance in the Xinjiang region of China (Chen et al., 2018b). These species exhibit remarkable adaptability to harsh conditions such as arid, saline–alkali, and infertile environments, leading to their widespread presence in major cities across the northwestern regions of China, including Xinjiang (Mao et al., 2022; Klich. 2000). The fruits of Russian olive are rich in sugars, proteins, fats, vitamins, and various essential amino acids required by the human body, positioning them as natural health foods (Gao et al., 2019). Moreover, Russian olive possesses substantial medicinal value (Hamidpour et al., 2017), with properties such as anti-inflammatory (Ahmadiani et al., 2000) and analgesic (Karimifar et al., 2017) properties, and has been used to treat conditions such as oral lichen planus (Beigom Taheri et al., 2010) and gastrointestinal ulcers (Gürbüz et al., 2003). In addition to their edible and medicinal uses, various parts of the Russian olive plant are utilized in the perfume industry, as well as in the production of wood products and musical instruments (Kiseleva and Chindyaeva. 2011). Although the germplasm resources of Russian olive in Xinjiang are well established and genetically diverse, genetic research on Russian olive lags behind that on other economic trees and field crops, failing to meet the growing industrial demands and the urgent need for variety improvement and cultivation. Prior research has focused primarily on the nutritional and medicinal component analysis of Russian olive fruits, introduction trials, functional component extraction, and pharmacological activity studies (Hassanzadeh and Hassanpour. 2018; Farzaei et al., 2015). The deficiency in genomic research on Russian olive severely limits the identification, utilization, and improvement of its germplasm resources.

During the prolonged processes of natural selection and artificial domestication, significant differentiation in fruit size has occurred in

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Russian olive, leading to the formation of two distinct fruit types: large-fruited (*Elaeagnus moorcroftii* Wall. ex Schlecht) and small-fruited (*Elaeagnus oxycarpa* Schltdl. or *Elaeagnus angustifolia* L.) (Ma et al., 2021). Although the taxonomic relationships among *E. moorcroftii, E. oxycarpa*, and *E. angustifolia* remain controversial (Huang and Maimaitijiang., 2005), previous studies have confirmed significant differences between *E. moorcroftii* and *E. oxycarpa* based on comprehensive morphological comparisons of leaves, flowers, fruits, and seeds, including traits such as fruit size, leaf base shape, leaf length, petiole length, style length, pedicel length, and anther length (Yu et al., 2012). Ma et al. (2021) further confirmed *E. moorcroftii* as a distinct species independent of *E. angustifolia* through systematic analyses of 16 morphological traits and 9 SSR markers. Gu et al. (2026), using 71 chloroplast genomes from the family Elaeagnaceae, revealed that *E. oxycarpa* and *E. angustifolia* cluster in the same clade with close genetic affinity. Collectively, these results indicate that *E. moorcroftii* is a separate species, whereas *E. oxycarpa* and *E. angustifolia* are relatively closely related. Fruit size is an important economic trait of flowering plants and directly affects the economic value of Russian olive (Zhao et al., 2021; Miranda et al., 2023). From an ontogenetic standpoint, the final fruit size is determined by a continuous process involving gynoecium formation, fertilization, and fruit growth (including cell proliferation, differentiation, and expansion) (Hussain et al., 2020; Tanksley. 2004). Research in *Arabidopsis* has shown that before fertilization, the construction of the gynoecium group's apical–basal, mediolateral, and abaxial–adaxial axes determines fruit length, width, and thickness, respectively (Tanksley. 2004). During the fruit growth phase after fertilization, the frequency of cell division and the duration of the cell cycle play significant roles in determining the final fruit size (Zhao et al., 2021; Hussain et al., 2020). Previous studies have indicated that the *SHATTERPROOF* gene participates in the development and dehiscence of *Arabidopsis* seeds by controlling the differentiation of the dehiscence zone (Ferrándiz et al., 2000). *SEEDSTICK (STK)* is essential for the development of the funiculus-like structure connecting the developing seed to fruit, regulating cell expansion and division, and loss-of-function mutants present shorter siliques (Pinyopich et al., 2003). Notably, plant hormones (including auxins, cytokinins, and ethylene) also participate in the regulation of fruit size by affecting the expression and transcription of genes (such as hormone synthesis-related genes) and cell division and growth (Hussain et al., 2020; Lacombe and Achard. 2016). For example, the heterologous expression of soybean *GmYUCCA5*, a key enzyme in IAA synthesis, in *Arabidopsis* resulted in fewer and shorter siliques (Wang et al., 2017). In wax gourd, small auxin-upregulated RNA (*SAUR*) is considered an important regulatory gene for large fruit size (Xie et al., 2019). Studies have also shown that by reducing the levels of ACC, the direct precursor of ethylene biosynthesis, transgenic canola plants expressing ACC-deaminase produce relatively small siliques and seeds (Walton et al., 2012).

In general, current research on the regulation of fruit size has focused primarily on model plants such as *Arabidopsis* (Wang et al., 2016), rice (Hao et al., 2021), wheat (Liu et al., 2023), and horticultural crops such as tomato (Yuste-Lisbona et al., 2020) and citrus (Zhang et al., 2023). Although fruit size is crucial for the yield of Russian olive, the complexity of its regulation is not well understood, partly owing to the limited genetic information obtained from Russian olive to date. With the reduction in sequencing costs and the significant improvement in sequencing efficiency, whole-genome resequencing (WGRS) technology has become one of the fastest and most effective methods for studying plant economic traits and performing molecular breeding (Yu et al., 2018). WGRS can effectively mine information on genetic sequence differences and structural variations, and when combined with phenotypic variation analysis, it can more accurately explain the reasons for the differentiation of fruit size in Russian olive populations. Compared with the small-fruited Russian olive (*E. oxycarpa*), the large-fruited Russian olive (*E. moorcroftii*) boasts superior taste, richer nutritional content, and higher yields, thus gaining greater popularity in the market (Wang et al., 2024a). In this study, a chromosome-level genome of *E. moorcroftii* was assembled, and 70 Russian olive accessions from six cities in the southern and northern parts of the Tianshan region of the Xinjiang Uyghur Autonomous Region, China, were resequenced. This study revealed the population structure of Russian olive in the Xinjiang region through whole-genome variation and phylogenetic analysis. Additionally, by combining selection sweep analysis with phenotypic variation, this study aimed to identify genetic variations associated with fruit size in Russian olive. This research reveals crucial genes involved in the genetic regulatory mechanisms of fruit size in Russian olive and contributes to the improvement of *Elaeagnus* varieties.

2. Materials and methods

2.1. Plant material and genome sequencing

The seeds of *E. moorcroftii* were collected in Urumqi, Xinjiang Province, China, and then planted in the botanical greenhouse of Xinjiang University (43.76,409°N, 87.61,586°E, 932.0 masl). Young leaves of mature *E. moorcroftii* plants (120 days old) were selected for whole-genome sequencing and *de novo* assembly. Seventy wild Russian olive samples, collected from six cities in the southern and northern regions of the Tianshan area, Xinjiang Uyghur Autonomous Region, China (including Altay, Changji, Jiashi, Shache, Yili, and Korla), were utilized for genome resequencing (Table S1). After collection, the samples were immediately immersed in liquid nitrogen and stored at −80 °C for subsequent DNA extraction. Genomic DNA was extracted using the QIAGEN Genomic Kit according to standard operating procedures. Paired-end Illumina, Nanopore 1D ligation, and high-throughput chromosome conformation capture (Hi-C) libraries were constructed for sequencing and sequenced on an Illumina NovaSeq 6000 platform (Illumina, San Diego, CA, USA), a Nanopore GridION X5 platform (Oxford Nanopore Technologies, Oxford Science Park, UK), and an Illumina NovaSeq 6000 platform, respectively.

2.2. Genome survey, assembly, and evaluation

We first performed filtering on short Illumina reads using fastp (Chen et al., 2018a), and K-mer analysis was employed to estimate the genome size of *E. moorcroftii* (Zerbino and Birney. 2008). The 17-mer frequency of the NovaSeq reads was determined using Jellyfish (v2.3.0) (Marçais and Kingsford. 2011).

The Oxford Nanopore Technologies (ONT) long reads were corrected and *de novo* assembled using NextDenovo (https://github.com/Nextomics/NextDenovo). Illumina sequencing reads were aligned to the genome assembled from ONT sequencing data using the default parameters of BWA MEM (Li and Durbin. 2009), and then, polish correction was performed with Pilon to obtain the final genome sequence (Walker et al., 2014). The preliminary genome was subsequently assembled into pseudochromosomes using HiC-Pro (Servant et al., 2015). LACHESIS (https://github.com/shendurelab/LACHESIS) was employed to cluster, reorder, and orientate the corrected contigs into pseudochromosomes (Burton et al., 2013). The completeness and quality of the genome assembly were evaluated by benchmarking universal single-copy orthologs (BUSCOs, v5.2.2) (Simao et al., 2015) and the long terminal repeat (LTR) assembly index (LAI) (Ou et al., 2018).

2.3. Genome annotation

To predict transposable elements, a combined approach of homology- and *de novo*-based annotation methods was employed. The repeat sequences were annotated using RepeatModeler (Flynn et al., 2020) and LTR-FINDER (Xu and Wang. 2007) based on *de novo* methods. The repeats from the similarity-based predictions were detected using RepeatMasker and RepeatProteinMask (Tarailo-Graovac and Chen.

2009). For the prediction of protein-coding genes, *de novo*, homology-based, and RNA-seq-based predictions were conducted. *De novo* prediction was performed with default parameters using Augustus (Stanke et al., 2006), Glimmerhmm (Majoros et al., 2004), SNAP, and GeneID (Alioto et al., 2018). For homology-based prediction, the protein sequences of *Arabidopsis thaliana, Cannabis sativa, Elaeagnus angustifolia, Hippophae rhamnoides*, and *Ziziphus jujuba* were downloaded and aligned against the *E. moorcroftii* genome using GeMoMa and Genewise (Birney and Durbin. 2000) with the default parameters. RNA-seq reads from tissues (leaves, roots, stems, and seeds) were mapped to the assembled genome using HISTA2 (v 2.1.0), and the result was subsequently used to construct the transcripts in Cufflinks. PASA was further utilized for RNA-seq-based gene prediction. Finally, EvidenceModeler (EVM, v1.1.1) (Haas et al., 2008) was employed to integrate the results of these three methods, yielding the final protein-coding gene set. The annotated genes of the assembled genome were evaluated using complete BUSCOs (v5.2.2) (Simao et al., 2015). For gene function annotation, the predicted genes were blasted against the SwissProt (https://www.expasy.org/resources/uniprotkb-swiss-prot), NCBI nonredundant (NR) (https://www.ncbi.nlm.nih.gov/refseq/about/nonredundantproteins/), and Clusters of Orthologous Groups for Eukaryotic complete Genomes (KOG) (http://genome.jgi-psf.org/help/kogbrowser.jsf) databases using BLASTP, with an E-value threshold of 1e-5 for optimal matches. Additionally, potential Gene Ontology (GO) terms and pathways for the predicted genes were obtained using blast2GO and Kyoto Encyclopedia of Genes and Genomes (KEGG) with default parameters (Conesa et al., 2005). For the annotation of different types of noncoding RNAs, tRNAscan-SE (v1.3.1) (Lowe and Eddy. 1997) and miRBase (v21) were used to predict tRNAs and miRNAs with eukaryotic parameters, respectively. The rRNAs were identified by Rfam (v12.0) (Griffiths-Jones et al., 2005), and the snRNAs were predicted using INFERNAL based on Rfam (v1.1).

2.4. Gene family, phylogenetic and whole-genome duplication (WGD) analyses

Genome sequences from the plant species *Amborella trichopoda, Arabidopsis thaliana, Elaeagnus mollis, Glycine* max, *Hippophae rhamnoides, Populus trichocarpa*, and *Ziziphus jujuba* were obtained, with the download addresses provided in Supplementary Table S10. OrthoFinder2 was utilized to identify orthologous genes between *E. moorcroftii* and the aforementioned seven species (Emms and Kelly. 2015). Single-copy genes from these species were employed to construct a phylogenetic tree of species evolution using RAxML (with a bootstrap setting of 100) (Stamatakis. 2006). Divergence times were estimated using the MCMCTree program within the PAML software package (Yang. 2007). To determine the expansion and contraction of gene families, the CAFÉ program (version 3.1) was utilized for analysis (De Bie et al., 2006). To identify whole-genome duplication events (WGDs), BLASTP software was employed (with an *E* value $\leq 1e^{-5}$) for reciprocal protein comparisons to identify conserved orthologous genes across the species (Li et al., 2003). The synonymous substitutions per synonymous site (*Ks*) values of the orthologous genes within these species were subsequently calculated using PAML, and the *Ks* values for *E. moorcroftii* and the other species were analyzed to predict WGD events.

2.5. Population genome resequencing

The genomic DNA (gDNA) of 70 wild *E. moorcroftii* and *E. oxycarpa* samples was extracted from fresh leaves. After the quality and concentration of the gDNA were assessed, gDNA sequencing libraries were constructed. The random shearing of DNA fragments of the desired length was subsequently performed by Covaris, followed by adapter ligation and cluster preparation using either the Solexa or E-PCR (SOLiD) methods. Finally, the inserted fragments were subjected to resequencing using the paired-end (Solexa) method. The sequencing was performed on an Illumina HiSeq with a PE150 sequencing protocol.

2.6. Variation calling

Fastp software (https://github.com/OpenGene/fastp) was used to filter the raw reads obtained from Illumina sequencing to remove low-quality reads with adapters. The obtained high-quality clean reads were subsequently mapped to a reference genome using BWA software (version 0.7.8). Duplicates were removed from the alignment results using SAMtools (version 0.1.19) (Li et al., 2009). SNPs and indels were detected using the HaplotypeCaller tool from GATK software (version 4.1.9.0) (McKenna et al., 2010). Low-frequency and low-quality SNPs were filtered out using VCFtools. Finally, the genomic regions and quantities of SNPs and indels were annotated using ANNOVAR (Wang et al., 2010) based on the GFF3 file and the reference genome of *E. moorcroftii*.

2.7. Population structure and genetic diversity analysis

Following variation calling, SNPs were filtered using VCFTools (Danecek et al., 2011) with parameters set to –min-alleles 2 –max-alleles 2 –min-meanDP 5 –maf 0.05 –max-missing 0.5. The filtered SNPs were then utilized for further analysis. Initially, population structure analysis was conducted using ADMIXTURE (version 1.23), which tests cross-validation errors for each clustering value (K) ranging from 2– to 9. A maximum likelihood (ML) phylogenetic tree was subsequently constructed using the IQ-TREE (v3.1.3). For principal component analysis (PCA), GCTA (version 1.24.2) was employed (Yang et al., 2013).

2.8. Selective sweep scanning

To identify genomic regions potentially affected by selection, two methods were employed in concert: the population fixation index (F_{ST}) and the reduction of diversity (ROD). Nucleotide diversity (π) and the F_{ST} were calculated using VCFtools (version 0.1.13, https://vcftools.github.io) within a 50-kb sliding window with a step size of 5 kb. The pairwise genomic differentiation among subpopulations for candidate genes was estimated using π and F_{ST}. The ROD values were calculated based on the ratio of π for a subpopulation relative to a control subpopulation. All output results for ROD and F_{ST} were standardized and converted into z scores using a 50-kb sliding window with a step size of 5 kb. Genomic regions were selected as candidates based on the top 5% of values.

2.9. Gene cloning and genetic transformation

Fresh *E. moorcroftii* leaves were collected and ground in liquid nitrogen. Total RNA was extracted using an E.Z.N.A. Plant RNA Kit (cat. no. R6827–01; Omega Bio-tek, USA). The concentration and purity of the extracted RNA were determined with a NanoDrop ND-1000 spectrophotometer (Thermo Fisher Scientific, USA). Subsequently, first-strand cDNA was synthesized using the One-Step gDNA Removal and cDNA Synthesis SuperMix (TransGen Biotech, Beijing, China).

To validate the regulatory roles of the identified candidate genes in fruit size control of Russian olive, we selected two genes with significant expression differences between *E. moorcroftii* and *E. oxycarpa* and two genes without significant differences for subsequent cloning and functional validation, based on RNA-seq data (Table S17, unpublished data). Using the synthesized cDNA as a template, the coding sequences of four genes lacking stop codons—*EmGATA1* (1026 bp), *EmIAA4* (273 bp), *EmTUBB6* (1350 bp), and *EmUGT74F2* (1425 bp)—were amplified via PCR and subsequently cloned into the pCAMBIA1302 vector. The primer sequences utilized are listed in Supplementary Table S1. The four recombinant constructs were individually introduced into *Arabidopsis* via

the Agrobacterium-mediated floral dip method (Clough and Bent, 1998). Harvested T_1 seeds were sown on a selective medium containing hygromycin to screen for resistant transgenic plants. Putative positive transformants (T_1 generation) were transplanted to soil for continued growth. After 4–5 weeks of cultivation, genomic DNA was extracted from the leaves and subjected to PCR verification using hygromycin-specific primers (Supplementary Table S18). Seeds collected from these PCR-confirmed positive plants were designated as the T_2 generation and dried at 26 °C. This selection process was repeated over successive generations until homozygous T_3 *Arabidopsis* lines were obtained, which were utilized for all subsequent experiments.

Seeds of both the T_3 transgenic lines and wild-type *Arabidopsis* were sown in potting mixture and cultivated in a controlled plant growth chamber (25 °C, a 16-h light/8-h dark photoperiod, and a light intensity of 10,000 lx). Phenotypic variations were photographically documented, and appropriate samples were collected for the determination of relevant physiological parameters.

Statistical analysis and data visualization were performed using SPSS (version 26.0; IBM Corp., Armonk, NY, USA) and GraphPad Prism (version 10.0; GraphPad Software, Boston, MA, USA). Statistical significance among different treatment groups was evaluated using a one-way analysis of variance (ANOVA) followed by Duncan's multiple range test, with $p < 0.05$ indicating significant differences.

3. Results

3.1. Assessing genome size and genome assembly analysis

To explore the genetic basis of the Russian olive, we attempted to construct a high-quality genome. Based on K-mer analysis, the genome of *E. moorcroftii* is estimated to be 530 Mb in size, with a heterozygosity of 0.71% (Figure S1). From Oxford Nanopore Technology (ONT) sequencing data, we obtained 46.3 Gb of clean reads with an average length of 21.42 kb and a contig N50 of 26.9 Mb and preliminarily assembled them into a 504 Mb genome. We subsequently used short reads from Illumina to correct the assembled genome, ultimately obtaining a 505 Mb genome, which included 46 contigs with a contig N50 of 27.42 Mb (Table S2).

To anchor the assembled contigs to presumed chromosomes, we constructed a Hi-C library for *E. moorcroftii*. A total of 46.72 Gb of clean reads were obtained (92x genome coverage), with a Q30 ratio greater than 92.1%. The results revealed 75.86 million unique read pairs aligned to the genome, of which 62.09 million were valid interaction pairs, accounting for 81.84% of the genomic data (Table S3). After high-throughput chromosome conformation capture (Hi-C)-assisted assembly, a total of 501,468,071 scaffolds were ordered and positioned onto 14 presumed chromosomes, with a mounting rate of 99.19%, a scaffold N50 of 30.3 Mb, and a maximum chromosome length of 103.1 Mb (Table S4). The final chromosome-level genome size of *E. moorcroftii* was determined to be 505.5 Mb (Table S5), with no apparent assembly errors, including 14 chromosomes, and the Hi-C interaction heatmap is shown in Fig. 1b

We further validated the quality of the assembled genome using BUSCO analysis. Among the 1375 plant-specific orthologs, 1328 (96.6%) were identified. The results indicated that the genome sequence has high continuity and sequence quality (Table S6). A total of 33,650 protein-coding genes and 297 miRNAs were functionally annotated using various methods, with an average gene length of 3474 bp, a coding sequence size of 1200 bp, and a mean of 5.27 exons. The *E. moorcroftii* genome presented similar lengths of genes, coding sequences (CDSs), exons, and introns as those of five other species, including *A. thaliana, C. sativa, E. angustifolia, H. rhamnoides*, and *Z. jujuba* (Figure S2). Further functional annotation revealed that 98.12% and 79.59% of the protein-coding genes were annotated in the NR and SwissProt databases, respectively. Additionally, 45.32% and 34.29% of these genes were classified and annotated by GO terms and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways, respectively (Table S7). By combining *de novo* annotation and homology prediction methods, a total of 256.2 Mb (50.68%) of the assembled sequences in the *E. moorcroftii* genome were identified as repeats, including 13.2 Mb of DNA transposons and 158.5 Mb of long terminal repeat (LTR) retrotransposons. LTR retrotransposons are the most abundant type of repetitive sequence in *E. moorcroftii*, accounting for 31.35% of the genomic sequence (Table S8). Moreover, 881 tRNAs, 1079 rRNAs, and 2931 snRNAs were annotated in the *E. moorcroftii* genome (Table S9).

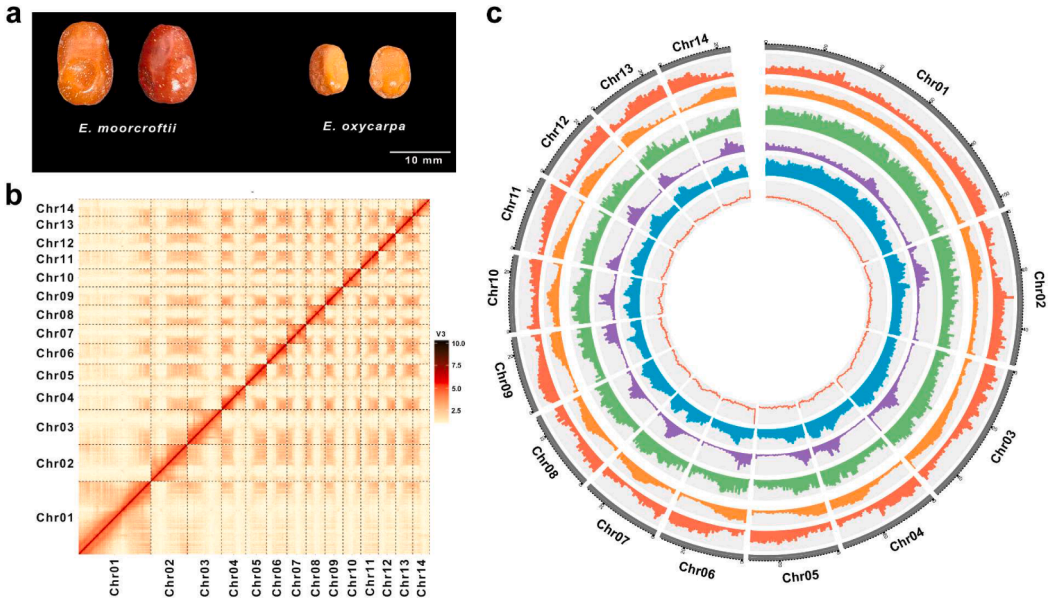


Fig. 1. Morphological characteristics of the fruit, Hi-C interaction heatmap, and chromosomal features of *E. moorcroftii*. (a) Morphological characteristics of fruit from *E. moorcroftii* and *E. oxycarpa*. (b) Hi-C interaction heatmap of *E. moorcroftii*. (c) Overview of the *E. moorcroftii* genome. The tracks from the outer to inner regions of the circle indicate chromosomes, the density of coding genes, long terminal repeats(LTRs), Copia retrotransposons, Gypsy retrotransposons, the density of SNPs, and the GC content in the *E. moorcroftii* genome.

3.2. Evolution and functional differentiation of the E. moorcroftii genome

To investigate the evolution and divergence of the *E. moorcroftii* genome, we employed OrthoMCL to compare the genomic relationships between *E. moorcroftii* and seven other sequenced plant species. A phylogenetic tree was constructed based on 461 single-copy orthologous genes common to the eight species (Fig. 2). The results of the phylogenetic analysis indicated that Russian olive is most closely related to *Elaeagnus mollis*, and both belong to the genus *Elaeagnus*, followed by *Hippophae rhamnoides*, which belongs to the same *Elaeagnaceae* family. Their divergence times were estimated to have occurred 4.96 and 17.35 million years ago, respectively. In the *E. moorcroftii* genome, a total of 32,165 genes were clustered into 15,308 gene families, with 938 family expansions and 482 contractions compared with those of the other seven species (Fig. 2 and Table S11). KEGG enrichment analysis revealed that the expanded gene families in *E. moorcroftii* were enriched primarily in metabolic processes such as photosynthesis, pyruvate metabolism, methane metabolism, and cysteine and methionine metabolism. GO enrichment analysis indicated that the expanded gene families were enriched mainly in photosynthesis, photosystem, and chloroplast (Figure S3 and Figure S4). To assess potential whole-genome duplication (WGD) events in the *E. moorcroftii* genome, the *Ks* distributions of representative Elaeagnaceae species (including *E. mollis, E. moorcroftii*, and *H. rhamnoides*) and *Z. jujuba* were calculated (Figure S5). The *Ks* peaks for *E. moorcroftii, E. mollis*, and *H. rhamnoides* all occurred in the range of 0.28–0.36, suggesting that they may have experienced a common genomic duplication event, which is consistent with the results of the phylogenetic relationships.

3.3. Population genome resequencing

We collected 70 individual Russian olive samples from six cites in Xinjiang that presented varying fruit sizes. The whole-genome resequencing data of the 70 Russian olive materials produced 616.09 Gb of high-quality clean bases, with an average of 96.60% of the reads having a base error rate < 1% (Q20) and 90.33% of the reads having a base error rate < 0.1% (Q30). The average sequencing coverage depth was 14 ×, with a genome coverage rate of 87.60% (Table S12). The alignment of the clean reads to the *E. moorcroftii* genome resulted in an average mapping rate of 85.8%. A total of 5428,784 high-quality SNPs and 745,758 indels were identified. To avoid the influence of SNPs in linkage disequilibrium with each other, we filtered using $r^2 = 0.8$, retaining 1385,284 SNP loci for subsequent analysis.

To investigate the genetic structure of the 70 Russian olive populations, a Bayesian clustering algorithm based on a mixed model was used to estimate the ancestry proportion of each individual. When K = 5, the error distribution of the K-number reached the lowest value, and the Russian olive population could be divided into five groups(Q1-Q5) (Fig. 3b and c). The geographic distribution map of sampling sites showed a clear correspondence between these five genetic clusters and the geographic locations of the six sampled cities (Figure S6), indicating that this clustering reflects the geographic distribution pattern of the populations. The phylogenetic tree analysis is largely congruent with the STRUCTURE and PCA results, with the exception of samples NO.48 (Jiashi small-fruited type), NO.52 (Korla small-fruited type), and NO.55 (Altay large-fruited type) as shown in Fig. 3d These three accessions exhibit admixed ancestry, with ancestry coefficients exceeding 10% in multiple genetic clusters, which may be attributable to historical hybridization or recent gene flow among geographically proximate populations.

To further quantify the effect of geographic factors on gene flow, we calculated pairwise F_{ST} values among the five subpopulations. The F_{ST} values ranged from 0.144 to 0.375, with a mean of 0.226, indicating moderate to high genetic differentiation among subpopulations (Table S13). Subpopulation pairs with greater geographic distance (e.g., Q3 vs. Q5, $F_{ST} = 0.375$) exhibited higher differentiation, while geographically proximate pairs (e.g., Q2 vs. Q5, $F_{ST} = 0.144$) showed relatively lower differentiation; this pattern was consistent with the geographic proximity observed in the STRUCTURE analysis. Overall, these results suggest that geographic distance may be an important factor limiting gene flow in this species, but the extent of its effect varies among different subpopulation pairs. For example, Q1 included some individuals from Jiashi and Shache, whereas Q3 consisted entirely of Jiashi individuals; Jiashi and Shache are geographically close, and the phylogenetic tree also placed Q1 and Q3 in the same clade.

3.4. Identification of key genes regulating fruit size

We categorized 70 individual Russian olive fruits into two groups based on fruit length and width: large-fruited (n = 33) and small-fruited (n = 37). Initially, we conducted selection sweep analysis by calculating the F_{ST} between the large- and small-fruited groups, identifying 1444 candidate genes in the top 5% windows (Fig. 4 and Table S14). These candidate genes were significantly enriched in GO terms related to fruit growth and development, including those involved in morphogenesis, growth and development, cellular development, and seed development. Genes with strong selection signals are considered the foundation for the differentiation of fruit size in Russian olive (Fig. 5). Among them, genes involved in cell wall and cell elongation included beta-ketoacyl-CoA synthase (*KCS*), tubulin beta-6 chain (*TUBB6), GDP dissociation inhibitor* (*GDI*), and phosphatidylinositol transfer protein 5 (*AIPT5*). Genes responsive to auxin and salicylic acid signals included UDP-

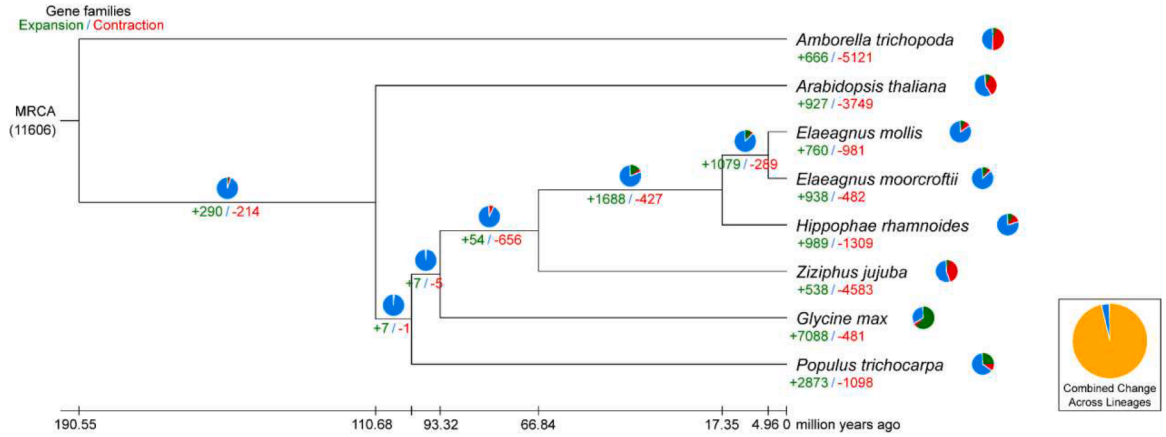


Fig. 2. Phylogenetic tree, divergence time, and expansion and contraction of gene families in eight species. Evolutionary relationships and divergence times were calculated based on 461 single-copy orthologous genes. Pie charts show the proportions of gene families that underwent expansion or contraction.

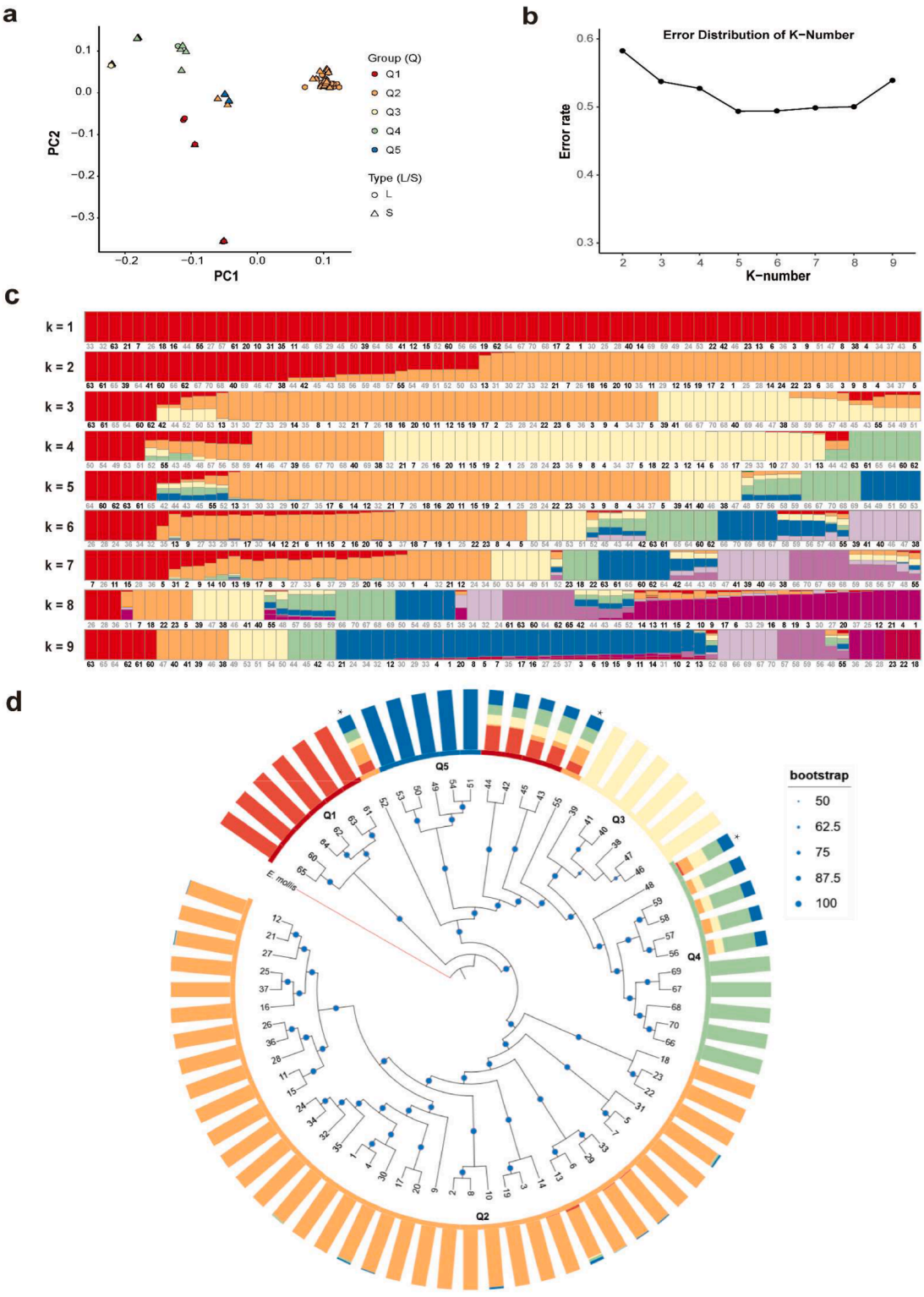


Fig. 3. PCA and population structure analysis of Russian olive in Xinjiang, China. (a) PCA plot of the first two eigenvectors of the Russian olive population. PC1 and PC2 divide the Russian olive populations into five groups, solid circles represent large-fruited accessions, and solid triangles represent small-fruited accessions. (b) Cross-validation error rates corresponding to different K values. (c) Population structure of Russian olive at cluster values (K) of 1–9. Each individual is denoted by a vertical bar composed of different colors corresponding to its proportion of genetic ancestry, Large-fruited accessions are labeled in black, and small-fruited accessions in gray. (d) Neighbor–joining (NJ) phylogenetic tree for 70 Russian olive individuals.

glycosyltransferase 74F2 (*UGT74F2*), auxin-responsive protein IAA4 (*IAA4*), PINOID-like kinase 2 (*PID2*), IAA-alanine hydrolase 3 (*IBR3*), serine/threonine-protein kinase (*STPK*), and guanine nucleotide-binding protein beta subunit (*GNB*). Genes playing key roles in cell cycle and cell division included AGAMOUS-like MADS-box protein 62 (*AGL62*), MEI2-like RNA-binding protein (*MEI2-like*), and GATA-type transcription factor (*GATA*).

To further identify key regulatory genes for fruit size in Russian

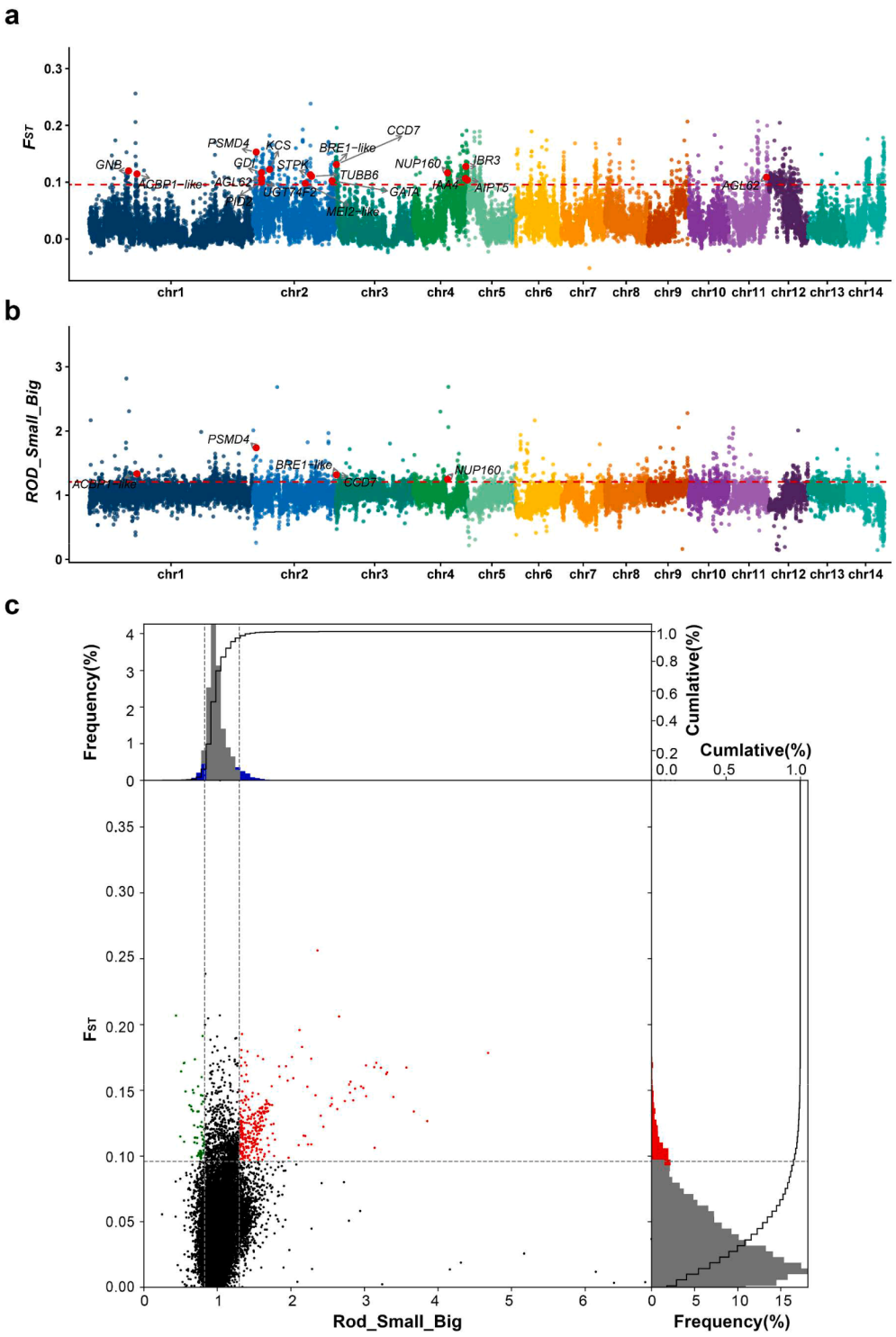


Fig. 4. Genome-wide screening of selected regions of *E. moorcroftii* and *E. oxycarpa*. (a) Genome-wide distribution of F_{ST}; (b) Genome-wide distribution of *ROD*; (c) Scatter plot of F_{ST} versus *ROD*. Red dashed lines indicate the top 5% thresholds, and red points denote candidate selective sweep regions.

olive, we used both F_{ST} and ROD to detect selective sweep signals (Table S15 and Fig. 4), thereby reducing false positives and improving the accuracy of the results. A total of 321 overlapping protein-coding genes were identified by the combined approach, including 26S proteasome non-ATPase regulatory subunit 4 (*PSMD4*), acyl-CoA-binding protein 1-like (*ACBP1-like*), carotenoid cleavage dioxygenase 7 (*CCD7*), nuclear pore complex protein 160 (*NUP160*), and E3 ubiquitin-protein ligase (*BRE1-like*). Among these, *BRE1-like* regulates chromatin state through epigenetic modifications, affecting the transcription of development-related genes, while *CCD7* is a key enzyme in the strigolactone biosynthesis pathway. Collectively, the two analyses yielded 19 key regulatory genes (Table S16), which may play important roles in the regulation of fruit size in Russian olive. (Table 1)

3.5. Characterization of key genes regulating fruit size

To investigate the potential involvement of the *E. angustifolia* genes *EmGATA1, EmIAA4, EmTUBB6*, and *EmUGT74F2* in fruit development,

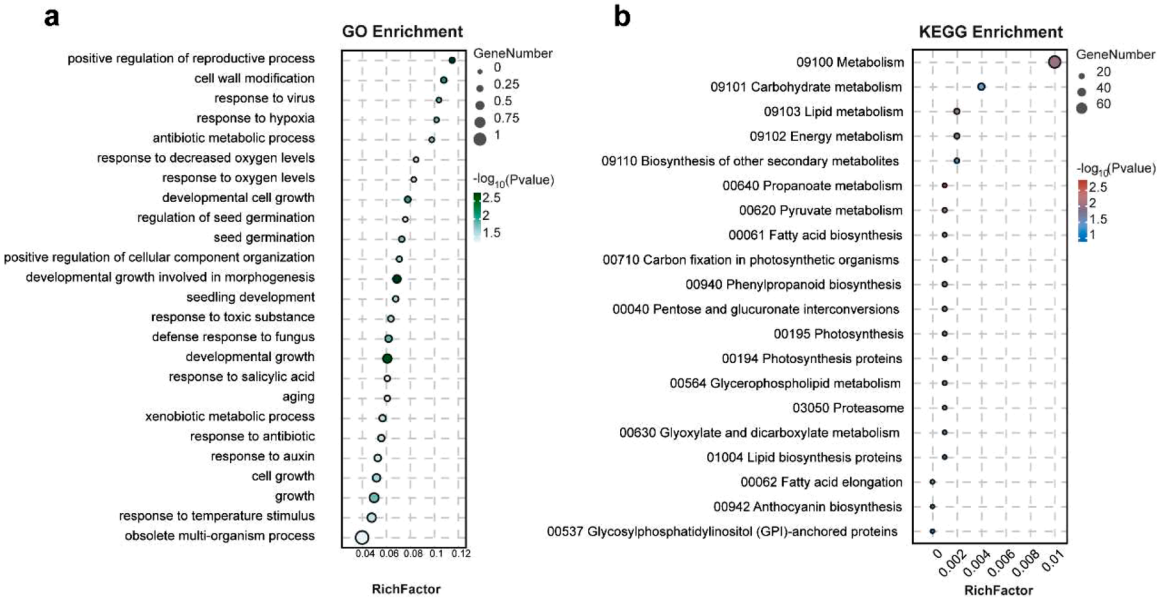


Fig. 5. Functional enrichment analysis of selective sweep candidate genes. (a) GO enrichment analysis of candidate genes for biological processes, where the size of the points represents the number of genes, and the color of the points indicates the significance level. (b) KEGG enrichment analysis of candidate genes, where the size of the points represents the number of genes, and the color of the points indicates the significance level.

Table 1
Summary of the *E. moorcroftii* genome assembly and annotation.

Category	Type	Length	Number
Assembly	Scaffolds	505.54 Mb	28
	Scaffold N50	30.33 Mb	5
	Scaffold N90	24.32 Mb	7
	Maximum length of scaffold	103.12 Mb	1
	Contigs	505.54 Mb	46
	Contig N50	27.42 Mb	5
	GC content	-	30.2%
	BUSCO	-	96.6%
Protein-coding genes	Total gene number	-	33,650
	Annotated gene	-	33,101
	Average length of CDS	1200.25 bp	-
	Mean exons per gene	227.7 bp	5.27

heterologous overexpression analysis was conducted via genetic transformation in *Arabidopsis* (Figure S7). The stably inherited T3 generation transgenic lines (*EmGATA1*-OE, *EmIAA4*-OE, *EmTUBB6*-OE, and *EmUGT74F2*-OE) and wild-type (WT) plants were cultivated under identical environmental conditions. Growth traits, including silique dimensions, plant height, and seed characteristics, were measured to evaluate the potential impact of the selected genes on fruit size and development (Fig. 6 and Table 2). After 60 days of growth, the transgenic plants exhibited significantly greater silique length, silique width, single-silique weight, and plant height compared to the WT. Specifically, the silique lengths of the *EmTUBB6*-OE, *EmIAA4*-OE, *EmGATA1*-OE, and *EmUGT74F2*-OE lines increased by 16.98%, 15.46%, 12.87%, and 12.72%, respectively, relative to the WT. On average, the silique width and single-silique weight of these four transgenic lines increased by 31.25% and 70.10%, respectively. Furthermore, the plant heights of the *EmGATA1*-OE, *EmTUBB6*-OE, *EmIAA4*-OE, and *EmUGT74F2*-OE lines were elevated by 45.26%, 39.02%, 32.82%, and 23.97%, respectively, compared to the WT. Taken together, the overexpression of *EmGATA1, EmIAA4, EmTUBB6*, and *EmUGT74F2* significantly promoted vegetative growth and silique development in *Arabidopsis*, which manifested as increased plant height, enlarged silique dimensions, and enhanced single-silique weight.

Subsequent morphological observation and quantitative analysis of mature seeds revealed no conspicuous differences in external appearance between the transgenic and WT seeds. However, measurements of the 200-seed weight demonstrated a significant increase in the mass of the transgenic seeds. Specifically, the 200-seed weights of the *EmUGT74F2*-OE, *EmGATA1*-OE, *EmIAA4*-OE, and *EmTUBB6*-OE lines increased by 25.17%, 17.88%, 6.62%, and 5.30%, respectively, compared to those of the WT (Table 2). These results indicate that although the external morphology remained unaltered, the overexpression of these genes significantly enhanced both the weight and plumpness of the seeds in the transgenic lines.

4. Discussion

Russian olive has significant economic and ecological value, especially in the southern Xinjiang region of China, where it serves as an important medicinal material and a raw material for the food industry (Waili et al., 2016; Ma et al., 2021). Generally speaking, fruit size can significantly affect the economic value of Russian olive (Wang et al., 2024b). The large-fruited variety (*E. moorcroftii*) contains higher levels of sugars, proteins, trace elements, and amino acids, with a better taste, making it the main research object for the breeding of superior varieties in the Russian olive industry (Wang et al., 2024a). However, current understanding of the regulatory mechanisms controlling Russian olive fruit size is still limited, and the lack of a high-quality Russian olive genome has severely hindered the in-depth analysis of its regulatory mechanisms. In this study, an integrated sequencing strategy was adopted, in which short reads generated by the Illumina NovaSeq platform were aligned to the genome assembled from third-generation sequencing (TGS) long reads to improve the accuracy of the final assembled genome. Meanwhile, Hi-C assisted assembly technology was used to anchor 501.49 Mb (99.19%) of the scaffold sequence onto 14 chromosomes. Although Fu et al. (2022) have published the first genome sequence of large-fruited *E. moorcroftii* using TGS, providing valuable genomic resources for Russian olive-related research, the long reads generated by TGS are prone to reducing sequencing accuracy, thereby affecting the quality of genome assembly. Compared with this genome, the genome assembled in this study showed increases of 4.63%, 7.40%, and 1.31% in anchoring rate, scaffold N50, and maximum chromosome length, respectively, which significantly improved the completeness of the genome.

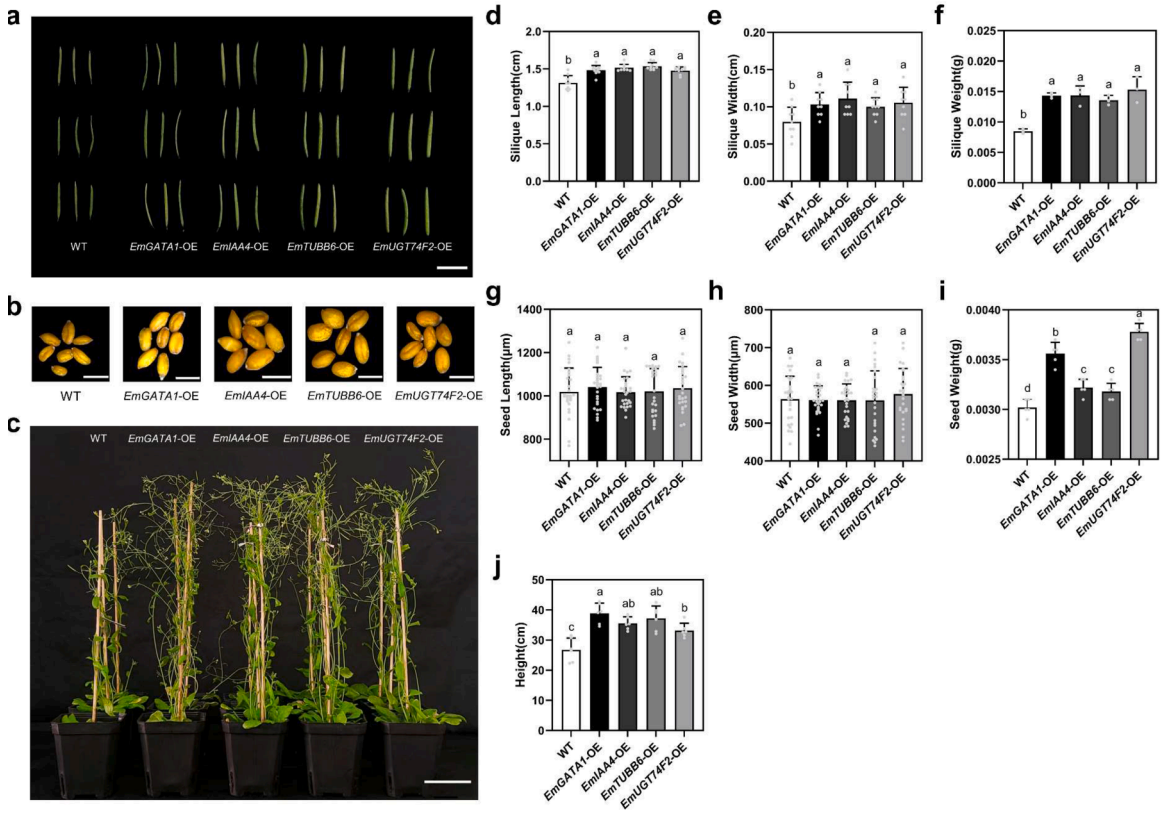


Fig. 6. Heterologous expression of *EmGATA1, EmIAA4, EmTUBB6*, and *EmUGT74F2* genes in *Arabidopsis thaliana* significantly increased silique and seed size. (a) Phenotypic observation of silique size. Bar=1 cm. (b) Phenotypic observation of seed size. Bar=1 mm. (c) Phenotypic observation of plant height. Bar=5 cm. (d), (e) and (f) Statistical comparison of silique length, width and weight in transgenic plants. (g), (h) and (i) Statistical comparison of seed length, width and weight in transgenic plants. (j) Statistical comparison of plant height in transgenic plants. Data are presented as mean ± standard error (n = 3). Different letters above the bars indicate significant differences at the $P < 0.05$ level.

Table 2
Measurement of indices in transgenic *Arabidopsis*.

Average value	WT	*EmGATA1*-OE	*EmIAA4*-OE	*EmTUBB6*-OE	*EmUGT74F2*-OE
Silique Length (cm)	1.31±0.10b	1.48±0.06a	1.52±0.05a	1.54±0.05a	1.48±0.05a
Silique Width (cm)	0.08±0.02b	0.10±0.02a	0.11±0.02a	0.10±0.01a	0.11±0.02a
Silique Weight (g)	0.0085±0.0004b	0.0143±0.0005a	0.0144±0.0016a	0.0136±0.0008a	0.0153±0.0021a
Seed Length (μm)	1019.00±109.40a	1040.22±91.52a	1017.11±71.29a	1021.15±105.08a	1035.96±99.78a
Seed Width (μm)	563.96±60.15a	561.15±37.54a	560.74±42.96a	560.33±78.02a	577.22±67.46a
Seed Weight (g, 200 seeds)	0.0030±0.0001d	0.0036±0.0001b	0.0032±0.0001c	0.0032±0.0001c	0.0038±0.0001a
Plant Height (cm)	26.78±3.90c	38.90±3.39a	35.57±2.15ab	37.23±4.08ab	33.20±2.40b

Data are presented as mean ± standard error (n = 3). Different letters within each row indicate significant differences at the $P < 0.05$ level.

E. moorcroftii (large-fruited Russian olive), together with *E. mollis* and *H. rhamnoides* (sea buckthorn), belongs to the Elaeagnaceae family and is a major ecological tree species in northwestern China (Cheng et al., 2020, 2022; Sevindik et al., 2023). Comparative genomic analysis showed that the divergence times of *E. moorcroftii* from *E. mollis* and *H. rhamnoides* are consistent with their taxonomic conclusions (Ferrer-Gallego. 2017). GO and KEGG enrichment analyses revealed that compared with their common ancestor, 938 expanded gene families in *E. moorcroftii* are mainly enriched in photosynthesis-related pathways, suggesting that *E. moorcroftii* may have undergone adaptive evolution related to photosynthesis. However, whether this adaptive evolution can promote the accumulation of photosynthetic assimilates and thereby drive fruit enlargement remains to be further explored.

Russian olive is native to Central Asia and has long been used in traditional medicine and as a food source in the Xinjiang region of China (Amiri Tehranizadeh et al., 2016). Throughout its long history of natural reproduction and artificial cultivation, it has accumulated rich genetic variation and has been subjected to extensive natural and anthropogenic selection. In six cities across the southern and northern parts of the Tianshan Mountains in Xinjiang, we collected 70 representative Russian olive accessions, including two varieties: large-fruited and small-fruited. Overall, phylogenetic tree, population structure, and principal component analyses revealed that the 70 Russian olive accessions can be divided into five groups. Our results show that the division of

populations is closely related to geographical factors. For example, Group 1 includes populations from the geographically close locations of Shache and Jiashi, whereas Group 4 consists of populations from Yili and Altay. Groups 2 and 5 are divided into two groups separated by the Tianshan Mountains. This finding is similar to that of a previous study on *Populus cathayana*, which was also influenced primarily by geographical factors such as climate and elevation, resulting in four groups in China (Xiang et al., 2023). However, in the five groups, large and small fruits were essentially evenly distributed (Table S1), suggesting that the size of Russian olive fruits is more susceptible to deliberate human interference than geographical factors. This may be because the variation in Russian olive fruit size is closely related to its economic value.

At the whole-genome level, selection sweep analysis can effectively identify beneficial genetic variations and improve breeding efficiency, which is crucial for breeding work. Lu et al. (2019) identified genes related to stress tolerance, seed quality, and content in 588 *Brassica napus* accessions through selection sweep analysis, providing important candidate targets for the genetic improvement of *B. napus*. Jiang et al. (2024) identified 1225 artificially selected genes in 185 tobacco samples, which can effectively distinguish between filler cigar tobacco and wrapper cigar tobacco. Fruit size directly determines the economic value of Russian olive, but to date, no genome-wide studies on the variation of Russian olive fruit size have been reported. In this study, through selection sweep analysis, we conducted the first genome-wide study on the regulation of Russian olive fruit size. Among the 1444 common genes identified in the top 5% F_{ST} windows of the large-fruited and small-fruited populations, many genes have been proven to be involved in cell division and cell cycle regulation, among which *AGL62, GATA1, IAA4, TUBB6, MEI2, GNB* and other genes are prominent in the genomic regions related to fruit size regulation. In *Arabidopsis thaliana*, the *MADS* domain protein *AGL62* can regulate floral organ and endosperm development (Kang et al., 2008), while *TUBB6* plays a key role in plant morphogenesis, determining the plane of cell division and the direction of cell expansion (Gavazzi et al., 2017), suggesting that these genes may play important regulatory roles in the fruit development of Russian olive.

In the fine regulation of fruit set, plant hormones are the core regulators that initiate the beginning of fruit set (Zhao et al., 2021). For example, auxin is involved in fruit formation and size by regulating cell division and expansion processes (Pattison and Catalá. 2012). In *Solanum lycopersicum* (tomato), silencing the *SlIAA7* gene results in a phenotype of larger fruits with thicker pericarp (Su et al., 2014); in *Oryza sativa* (rice), *UTG83A1* (also known as *GSA1*) is regulated by auxin levels and modulates grain size by controlling cell proliferation and expansion (Dong et al., 2020). In this study, 12 auxin-responsive genes were also identified, including the auxin-responsive protein gene *IAA4-like* and the UDP-glycosyltransferase gene *UGT74F2*, which have been proven to be involved in plant fruit development (Su et al., 2015; Wang et al., 2021).

The combined analysis of F_{ST} and ROD has been proven to be an effective method for detecting selection sweep regions (Lu et al., 2019). In this study, this method was used to further identify key genes regulating Russian olive fruit size, including *AGL62, IAA4, GATA, TUBB6*, and *UGT74F2*, which are speculated to play important roles in the regulation of Russian olive fruit development. To verify the functions of the above genes, heterologous expression was performed in *A. thaliana*. The results showed that the plant height and silique-related traits (length, width, and weight) of all transgenic lines achieved coordinated improvement. This phenotype may be closely related to the known or potential functions of the target genes in regulating plant cell elongation, division, and organ development.

5. Conclusion

In summary, we report a chromosome-level genome of the large-fruited Russian olive (*E. moorcroftii*) assembled using a combination of second-generation sequencing, third-generation sequencing, and Hi-C technology. We subsequently resequenced 70 samples of Russian olive, including large- and small-fruited varieties, collected from the Xinjiang region. Population structure analysis revealed that the Russian olive populations from six areas north and south of the Tianshan Mountains in Xinjiang can be divided into five groups. Additionally, we preliminarily explored the genetic basis of fruit size variation in Russian olive through selection sweep analysis. Our results provide new insights into the genomic variation, phylogenetic relationships and utilization of Russian olive, identify 19 key genes associated with fruit size regulation, and functionally validate four of these genes via heterologous expression in transgenic *Arabidopsis*. These results may enhance our understanding of the variation in fruit size in Russian olive and provide valuable genetic resources for its improvement.

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CRediT authorship contribution statement

Juan Zhao: Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Data curation. **Yueying Hu:** Writing – review & editing, Methodology, Investigation, Data curation. **Jing Zhang:** Writing – original draft, Software, Resources, Data curation. **Yuanyuan Wang:** Software, Methodology, Formal analysis, Data curation. **Qinghong Luo:** Resources, Project administration, Methodology, Data curation. **Sheng Zhang:** Writing – review & editing, Supervision, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.scienta.2026.115130.

Data availability

The genome assembly and annotation files, as well as the whole-genome resequencing data generated in this study, have been deposited in the CNSA (CNCB Nucleotide Sequence Archive) database under the project accession number [CNP0009945]. The data are publicly accessible at https://db.cngb.org/cnsa/.

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